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sala2code.github.io/portfolio/

Sala2Code

Toulouse, France

SALA RAPHAËL

SUMMARY

I am **SALA Raphaël**, a *Master's student in Applied Mathematics* at the University of Toulouse, preparing to pursue a PhD in mathematics. I am mainly interested in **probability and statistics**, with a particular interest in random structures and the inference problems associated with them.

Having studied **mathematics, physics, and computer science**, I have developed a broad scientific background that complements my current focus on theoretical mathematics. My internships have shaped my research interests, and I remain keen to pursue opportunities abroad.

PUBLICATION

- **Discretization-free exact recovery in geometric community detection.** *Raphaël Sala, Moritz Otto, Maarten Hoeneveld.* arXiv:2609:01635 (2026)

MATHEMATICAL PROJECTS

- **Community Detection in Geometric Random Graphs** Models, recovery thresholds, and inference methods.
- **Finite-Sample Analysis of the Λ -Coalescent TMRCA**, identifiability, and population genetics.
- **Spectral Community Detection in the SBM** Spectral clustering via matrix perturbation and random matrices.
- **Uniform Polynomial Approximation** Bernstein, Tchebychev, and minimax approximation.

OTHERS

- ANITI Graduate-School Excellence Scholarship (merit-based, two-time recipient: 2025, 2026)
- Annual projects supervised by a researcher
- Computer science projects (see Portfolio)
- Homework tutoring in middle schools

WORK EXPERIENCE

- Internship — **Leiden University**, Mathematical Institute (Netherlands)
May - July 2026 (3 months)
Supervisor: Moritz Otto

Research on exact community recovery in geometric random graphs. Developed and proved a discretization-free label propagation framework.

- Internship — **University of Geneva**, Section of Mathematics (Switzerland)
June - August 2025 (3 months)
Supervisor: Sylvain Sardy

Introduced a Quantile Universal Threshold strategy for automatic λ -selection in sparse learning, extending to non-convex penalties and shallow neural networks.

- Internship — **LAAS-CNRS** (France)
June - August 2024 (3 months)
Supervisors: Carole Rossi, Matthieu Jonckheere, Alain Esteve

Explored the design space of energetic nano-materials, using active learning and developed new metrics to evaluate the performance of Gaussian process predictive models.

- Internship — **Institut de Mathématiques de Toulouse** (France)
June - August 2023 (2 months)
Supervisor: Fabrice Deluzet

Developed and optimized numerical simulation tools for magnetized plasma physics studies, benchmarking them with Python, MATLAB and Julia (*Dynamic Languages*).

EDUCATION

- **Interactions between Computer Science and Mathematics for AI - Master degree**
University of Toulouse (2025-2027)
Mathematics - Computer Science
- **Selective University Program - Joint degree**
University of Toulouse (2022-2025)
Mathematics - Physics